

Equivalent Circuit-Based Analysis of Current Sharing in Planar Transformer Parallel Windings

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Abstract—Planar transformers with PCB windings are increasingly popular in modern power conversion systems due to their controllable design and high power density. The small core volume of planar transformers benefits from high-frequency operation. However, high-frequency operation also introduces undesirable effects such as current redistribution within the windings—specifically, induced eddy currents within individual winding layers and uneven current sharing among parallel layers. These phenomena significantly increase conduction losses at high frequencies. Importantly, a tradeoff exists between the eddy current effect and the current-sharing effect, as each dominates in different frequency ranges. This frequency dependence allows for the possibility of selecting an optimal winding design for a given operating frequency. To mitigate eddy current losses, interleaving the primary and secondary windings is commonly employed to minimize the magnetic field penetration through the conductors. Despite its importance, current sharing among parallel winding layers is often addressed through empirical design, finite element analysis (FEA), or complex mathematical modeling—all of which can be time-consuming and lack clear physical insight. To address this limitation, this paper presents a simple analytical model for calculating current distribution and conduction losses in parallel windings, based on the electrical equivalent circuit of the transformer. The proposed model not only facilitates efficient evaluation of winding configurations but also provides valuable physical insight into the mechanisms governing current sharing in planar transformers.

Keywords—parallel winding, planar transformer, current sharing, winding loss

NOMENCLATURE

$\mathbf{H}$	- magnetic field intensity (vector)
$\mathbf{B}$	- magnetic flux density (vector)
$\mathbf{E}$	- electric field intensity (vector)
$\mathbf{J}$	- current density (vector)
$\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z$	- unit vector of the rectangular coordinate axes
x, y, z	- rectangular coordinate axes (Fig. 1)
r, ϕ, z	- cylindrical coordinate axes
H_y	- magnitude of magnetic field intensity on the y -axis
J_x	- magnitude of current density on the x -axis
I	- magnitude of current

J	- magnitude of current density
H	- magnitude of magnetic field intensity
n	- number of layers
I_{pi}, I_{si}	- magnitude of current in the i th layer of the primary winding, the secondary winding
h	- thickness of a winding layer (Fig. 1)
R	- radius of the conductor (Fig. 3, Fig. 4)
Δh	- distance between 2 winding layers (Fig. 1)
c	- width of a winding layer (Fig. 1)
l	- mean length of a winding layer
R_{pi}, R_{si}	- resistance of the i th layer of primary winding, secondary winding
R_{ac}	- ac resistance
R_{dc}	- dc resistance
L_{pi}, L_{si}	- leakage inductance of the i th layer of the primary winding, the secondary winding
$L_{self.i}$	- self-inductance of the i th layer
M_{ij}	- mutual inductance between the i th and j th layer
σ, ρ	- ohmic conductivity, resistivity
μ	- permeability
μ_0	- vacuum permeability
ω	- angular frequency
f	- frequency
t	- time
$i(t)$	- instantaneous current
j	- $\sqrt{-1}$
P_c	- conduction loss
P_M	- complex magnetic power
δ	- skin depth $\sqrt{2/\omega\mu_0\sigma}$
γ	- propagation constant $(1 + j)/\delta$
∇	- nabla operator

I. INTRODUCTION

The rapid advancements in electric vehicles, consumer electronics, and data centers are driving the need for high-

density, high-efficiency power delivery solutions. A critical component in these systems is the isolated DC–DC converter, which uses a transformer to achieve effective voltage conversion [12]–[16]. However, transformer design plays a crucial role in the overall system, as transformers often occupy a large portion of the footprint and contribute substantially to power loss.

By employing PCB windings, transformers can adopt a low-profile, planar form factor compared to traditional wire-wound structures. As the development of the wide bandgap devices, higher operating frequencies are now feasible, enabling the use of smaller magnetic cores as the reduced volt-seconds. However, the rapidly changing magnetic field at high frequencies penetrating the windings induces additional eddy currents in the conductors, manifesting as skin effect and proximity effect, which significantly increase total winding loss [1]–[4], [11] [14]. The ac winding loss of a planar transformer with series-connected windings has been analytically derived from Maxwell's equations under one-dimensional (1-D) assumptions [1] [2]. These analytical models reveal that the number of adjacent current-carrying conductors substantially influences the winding loss at high frequencies due to the proximity effect.

Fig. 1(a) illustrates a 2:2 transformer with two primary winding layers (P1, P2) and two secondary winding layers (S1, S2) connected in series, forming a 2P2S configuration. When windings carrying the same current are closely spaced, the magnetic field intensity increases due to accumulation, as shown in the FEA results in Fig. 1(c). Consequently, significant eddy currents are induced, particularly in the inner layers (P1 and S2), as depicted in Fig. 1(b). Quantitatively, at 500 kHz, the outer layers (P2 and S1) exhibit an 18.6% increase in loss compared to their DC losses, while the inner layers (P1 and S2) suffer a 53.5% increase due to stronger proximity effect. To mitigate this effect, interleaving primary and secondary windings is preferred to reduce the peak magnetic field cutting through the windings and thus lower the AC losses [4]. Fig. 2(a) and 2(e) illustrate two interleaved structures of the 2P2S series winding configuration. As shown in Fig. 2(c) and 2(g), without field accumulation, less eddy current is induced in the interleaved structures, as seen in Figs. 2(b) and 2(f), compared to the non-interleaved case.

Due to the effects of eddy currents in transformer windings, increasing copper thickness becomes less effective for handling high AC current. Instead, multiple winding layers are connected in parallel to meet high power demands [10]. However, current sharing among these parallel layers is strongly influenced by the excitation frequency. Fig. 3(a) and 3(e) illustrate two interleaved 2P2S structures in parallel for a 1:1 transformer. Case 1, shown in Fig. 3(a), achieves more uniform current sharing across parallel layers, as indicated by the simulation results in Fig. 3(b). In contrast, Case 2, shown in Fig. 3(e), exhibits an uneven current sharing, as seen in Fig. 3(f). Due to its superior current balancing, Case 1 is commonly adopted in high-current transformer designs where multiple winding layers are used in parallel to ensure even current distribution [15], [16]. On the other hand, despite the uneven current sharing in Case 2, it may yield lower overall winding loss at higher frequencies. This is primarily due to the reduced eddy current generation in the inner winding layers. Specifically, Fig. 3(f) shows lower eddy current concentrations in the inner layers (S1 and P2) of Case 2,

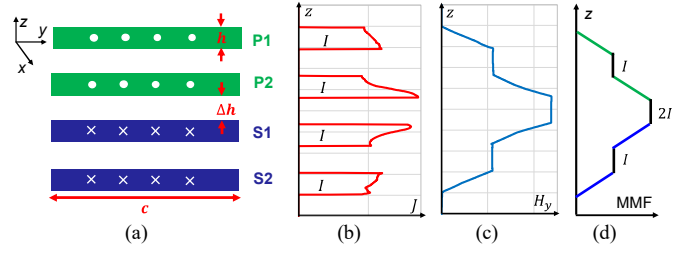


Fig. 1. (a) A non-interleaved 2P2S series winding structure of a 2:2 transformer and coordinate definition, (b) Simulated J , (c) Simulated H_y ($h = 3\text{oz}$, $\Delta h = 0.127\text{mm}$, $c = 1.8\text{mm}$, core diameter $d = 2.6\text{mm}$, $f = 500\text{kHz}$), (d) MMF graph.

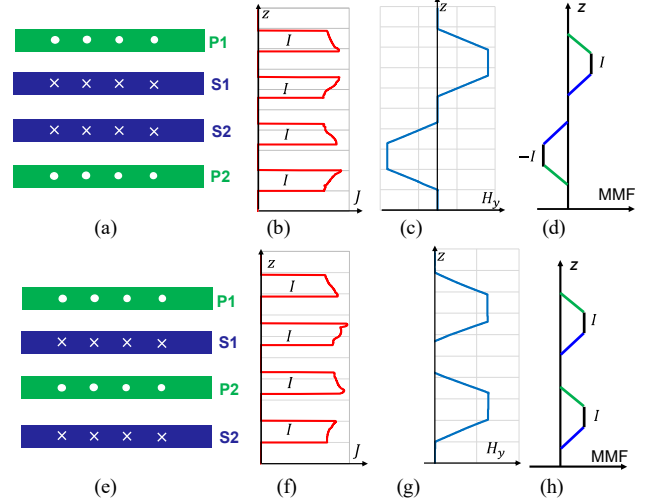


Fig. 2. Interleaved 2P2S series winding structures of a 2:2 transformer (a) Case 1, (b) Simulated J of Case 1, (c) Simulated H_y of Case 1, (d) MMF graph of Case 1, (e) Case 2, (f) Simulated J of Case 2, (g) Simulated H_y of Case 2, (h) MMF graph of Case 2 ($h = 3\text{oz}$, $\Delta h = 0.127\text{mm}$, $c = 1.8\text{mm}$, $d = 2.6\text{mm}$).

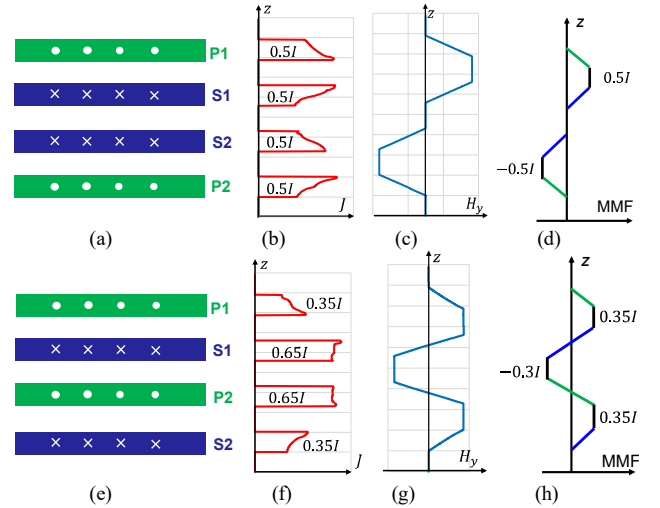


Fig. 3. Interleaved 2P2S parallel winding structures of a 1:1 transformer (a) Case 1, (b) Simulated J of Case 1, (c) Simulated H_y of Case 1, (d) MMF graph of Case 1, (e) Case 2, (f) Simulated J of Case 2, (g) Simulated H_y of Case 2, (h) MMF graph of Case 2 ($h = 3\text{oz}$, $\Delta h = 0.127\text{mm}$, $c = 1.8\text{mm}$, $d = 2.6\text{mm}$, $f = 1\text{MHz}$).

compared to the more pronounced eddy currents seen in Case 1 (as shown previously in Fig. 3(b)). This tradeoff highlights the importance of selecting an appropriate winding structure based on the operating frequency and dominant loss mechanisms.

Numerous studies have focused on modeling the current distribution between parallel winding layers. Finite element analysis (FEA) is commonly used to evaluate different parallel winding arrangements [8], but it can be computationally intensive when accounting for a variety of structures and design parameters. Analytical methods based on solving second-order differential equations derived from Maxwell's equations have also been proposed [5]-[7]. Yet, the complexity of these nonlinear equations often prevents closed-form solutions, particularly in cases involving multiple layers of primary and secondary windings. As a result, such approaches typically require structure-specific parameters and are limited to numerical solutions. Alternatively, circuit-based models can offer greater physical insight [17], [18]. A three-terminal impedance network is employed to model each winding layer in [9]. With this lumped circuit model, both current distribution and ac winding losses can be evaluated. However, the equivalent circuit presented in [9] becomes increasingly complex in multi-winding systems and is difficult to reconcile with conventional transformer modeling techniques [19], thereby limiting its practical applicability.

To address these challenges, this paper proposes a simple analytical solution for current sharing among parallel windings, based on the transformer T-model. This circuit-based model not only enables efficient evaluation of conduction loss but also offers physical insight into the current sharing mechanisms among parallel layers. Specifically, Section II introduces the frequency-dependent current redistribution effects within individual and parallel winding layers. Section III presents the equivalent circuit model and the corresponding analytical solution for current sharing. Moreover, building upon the 1-D analytical ac winding loss model for series-connected windings [2], Section III extends the analysis to derive the ac winding loss model for parallel windings. Winding loss of the two interleaved structures is evaluated at different frequencies, helping the optimal parallel winding design. Section IV concludes the paper.

II. FREQUENCY-RELATED CURRENT REDISTRIBUTION AND PARALLEL WINDING ANALYSIS

For an alternating current $i(t)$ through a conductor in Fig. 4, a time-varying magnetic field is generated in accordance with Ampère's law (1). This changing magnetic field induces an electric field $\mathbf{E}$ within the conductor, as described by Faraday's law (2) [21]. Inside the conductor, a linear relationship between the magnetic field intensity $\mathbf{H}$ and the magnetic flux density $\mathbf{B}$ is assumed through the vacuum permeability μ_0 (3). According to Ohm's law (4), the induced electric field drives an eddy current within the conductor.

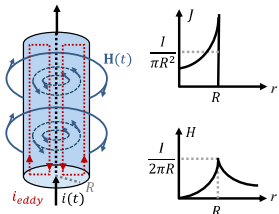


Fig. 4. Skin effect of a single conductor and the radial distribution of J , H .

$$\nabla \times \mathbf{H} = \mathbf{J} \quad (1)$$

$$\nabla \times \mathbf{E} = -\frac{d\mathbf{B}}{dt} \quad (2)$$

$$\mathbf{B} = \mu_0 \mathbf{H} \quad (3)$$

$$\mathbf{J} = \sigma \mathbf{E} \quad (4)$$

By combining equations (1)-(4), the relationship between the current density $\mathbf{J}$ and the

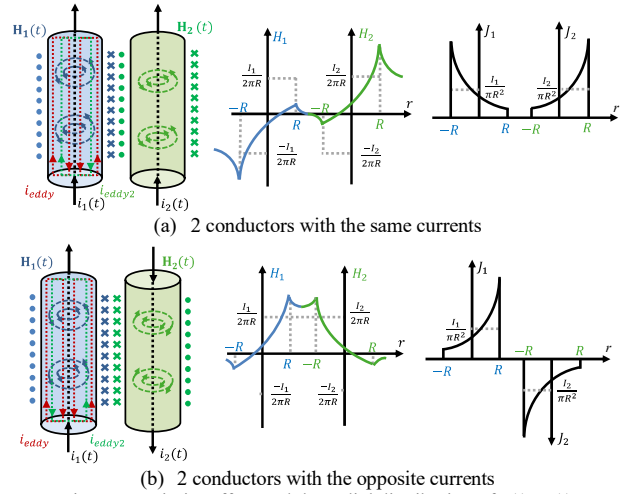


Fig. 5. Proximity effect and the radial distribution of $J(t)$, $H(t)$.

magnetic field intensity is derived in (5).

$$\frac{1}{\sigma} \nabla \times \mathbf{J} = -\mu_0 \frac{d\mathbf{H}}{dt} \quad (5)$$

From (5), two key insights can be drawn:

1. The higher the excitation frequency, the more eddy currents will be induced.
2. The higher the magnitude of magnetic field intensity, the more current crowding.

The eddy current effect on a single current-carrying conductor is illustrated in Fig. 4 [11]. The induced eddy current, shown as the red dashed line, reinforces the original current near the surface of the conductor while partially canceling it in the center. This results in an uneven current distribution within the conductor, known as the skin effect.

An extra frequency-dependent eddy current phenomenon is the proximity effect, which arises from time-varying magnetic fields generated by nearby current-carrying conductors [1], [2], [11]. For example, as shown in Fig. 5, when a second conductor carrying AC current is placed near the first, the changing magnetic field produced by the second conductor cuts through the first. This induces an eddy current—illustrated by the green dashed line—which further distorts the current distribution within the first conductor. The proximity effect, when combined with the self-field-induced skin effect, amplifies current crowding near the conductor surface and leads to increased conduction loss. From the simple case of current-carrying wires, an important observation can be made: the location of maximum current crowding coincides with the region of strongest magnetic field; the high magnetic field should be avoided.

For planar winding structures, the “MMF diagrams” [1], [11] is generally adopted to predict the current information. The scalar magnetic potential MMF (magnetomotive force), with its convenient unit, ampere turns, becomes an indicator of the magnetic field intensity. Several assumptions have been made for the 1-D field solution of planar transformers with the rectangular coordinate definition in Fig. 1(a):

1. The winding width is far greater than the winding thickness, and the winding is filling the winding window, such that the magnetic field is considered to be parallel to the winding surface and only varies along the vertical direction.

$$\mathbf{H} = H_y \mathbf{a}_y \quad (6)$$

$$\frac{\partial H_y}{\partial y} = 0 \quad (7)$$

2. The transformer core material has high permeability and no air gap, thereby the magnetic field intensity of the core is close to zero.

3. For power transformers, displacement current is far less than the electric current, so the capacitive effects can be ignored.

$$\mathbf{E} = E_x \mathbf{a}_x \quad (8)$$

Fig. 1(d) illustrates the distribution of MMF along the vertical axis of a 2:2 transformer with 2P2S non-interleaved windings in series. This MMF profile closely resembles the simulated magnetic field intensity shown in Fig. 1(c). As indicated, higher eddy currents are expected in the S2 and P1 layers due to the elevated peak MMF at these locations. To mitigate eddy current-related losses, it is essential to avoid high MMF peaks within the winding stack. For example, interleaved winding structures such as Case 1 in Fig. 2(a) with the sequence P-S-S-P and Case 2 in Fig. 2(e) with the sequence P-S-P-S exhibit reduced MMF peaks, as depicted in Fig. 2(d) and Fig. 2(h), respectively. These lower MMF peaks correspond to lower expected winding losses at high frequencies. Although the spatial distributions of MMF differ between Case 1 and Case 2, their peak MMF magnitudes are comparable. Therefore, their winding losses are expected to be similar. In summary, the shape and peak value of the MMF profile offer valuable insight into expected winding losses, providing a useful design metric in the early stages of planar transformer.

Due to eddy current effects, current tends to concentrate in regions with higher MMF. To accommodate high AC currents while minimizing loss, multiple winding layers—each thinner than the skin depth—are typically arranged in parallel and interleaved. However, an additional frequency-dependent phenomenon arises in parallel windings: non-uniform current sharing among layers [3]. Fig. 3 presents the current distribution in a 1:1 transformer with 2 layers of primary and 2 layers of secondary windings (2P2S) in parallel, for two interleaved configurations. Based on FEA results, Case 1 exhibits uniform current sharing across parallel layers at 1 MHz, as shown in Fig. 3(b). In contrast, Case 2 displays unequal current sharing, as seen in Fig. 3(f). Interestingly, despite this imbalance, Case 2 has a lower peak MMF than Case 1 (see Fig. 3(d) and Fig. 3(h)), which leads to reduced eddy current induction and a more uniform current distribution within individual layers. The total winding loss versus frequency for both cases is presented in Fig. 6. The results highlight two competing current redistribution mechanisms:

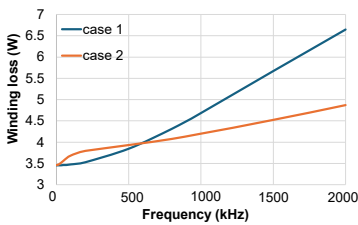


Fig. 6. Simulated winding loss of Case 1 & Case 2 of the 2P2S 1:1 transformer.

1. eddy current effects within each layer, and 2. frequency-dependent current sharing among parallel layers. These mechanisms dominate in different frequency regions. Specifically, the even inter-layer current sharing in Case 1 results in lower losses below 550 kHz, while reduced eddy currents in Case 2 make it more efficient above 550 kHz. Therefore, an optimal parallel winding design must account for both eddy current effects and inter-layer current sharing dynamics, particularly under varying frequency conditions.

III. CIRCUIT-BASED MODEL FOR CURRENT SHARING BETWEEN PARALLEL WINDINGS AND WINDING LOSS

To determine an optimal parallel winding arrangement, an analytical solution for parallel windings will be discussed in this section. Previously, a 1-D winding loss model has been derived in [2]. With the 1-D assumptions (6)-(8), and sinusoidal excitation, the differential form of Maxwell equations (1), (2) can be expressed as

$$-\frac{\partial H_y}{\partial z} = J_x \quad (9)$$

$$\frac{\partial J_x}{\partial z} = -j\omega\sigma\mu H_y \quad (10)$$

The magnetic field intensity and current density along the z -axis can be derived from (9), (8).

$$H_y(z) = C_1 e^{\gamma z} + C_2 e^{-\gamma z} \quad (11)$$

$$J_x(z) = -\gamma C_1 e^{\gamma z} + \gamma C_2 e^{-\gamma z} \quad (12)$$

Where $\begin{cases} C_1 = -\frac{1}{2}(H_y(0) - e^{h\gamma} H_y(h))(\coth h\gamma - 1) \\ C_2 = \frac{1}{2}e^{h\gamma}(e^{h\gamma} H_y(0) - H_y(h))(\coth h\gamma - 1) \end{cases}$, $\gamma = \sqrt{j\sigma\omega\mu_0}$ is the propagation constant, $H_y(0), H_y(h)$ is the boundary magnetic field intensity on the bottom and top of the winding, respectively. From (12), the ac conduction loss of a single winding layer can be expressed as (13)

$$P_c = \frac{1}{2} \rho l c \int |J_x(z)|^2 dz \quad (13)$$

where l is the mean length of a winding layer, c is the winding width, h is the winding thickness. From (13), the ac conduction loss only depends on the current distribution along the z -axis, as in (12) for a given winding structure. In other words, if the boundary magnetic field intensity is known, the analytical conduction loss can be solved. For instance, based on the 1-D assumptions (6) and Ampere's law (1), the boundary magnetic field intensity of Case 2 in Fig. 3(e) is listed in Table I, where I_{pi}, I_{si} are the unknown currents carried by the i th layer of primary and secondary, respectively. Therefore, for an analytical conduction loss, the current carried by each layer of parallel windings should be solved.

For a 2-layer parallel winding connected with a DC voltage source and load resistance in Fig. 7, the current flowing through each layer is determined by the resistance of each layer. With this simple idea, ac current carried by each layer of parallel windings is

TABLE I. BOUNDARY MAGNETIC FIELD INTENSITY OF CASE 2.

	$H_y(0)$	$H_y(h)$
P1	$\frac{I_{p1}}{c}$	0
S1	$\frac{I_{p1} - I_{s1}}{c}$	$\frac{I_{p1}}{c}$
P2	$\frac{I_{s2}}{c}$	$\frac{I_{s2} - I_{p2}}{c}$
S2	0	$\frac{I_{s2}}{c}$

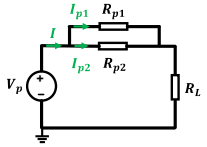


Fig. 7. Circuit of a 2-layer parallel winding connected with a voltage source and a load

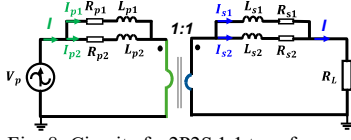


Fig. 8. Circuit of a 2P2S 1:1 transformer.

related to the impedance of each layer, including the resistance plus the inductance. Specifically, the circuit model for a 1:1 transformer a 2P2S winding in parallel is shown in Fig. 8 based on the transformer T-model [19], where the magnetizing inductance is assumed to be infinite and ignorable.

From the circuit model, the key to the current sharing of parallel windings is the impedance of each layer. For example, the current sharing for the 2P2S 1:1

transformer in Fig. 8 can be derived as

$$I_{p1} = \frac{Z_{p1} \parallel Z_{p2}}{Z_{p1}} I \quad (14)$$

where the layer impedance $Z_{p1} = R_{p1} + j\omega L_{p1}$, $Z_{p2} = R_{p2} + j\omega L_{p2}$. It can be seen that the layer resistance determines the current sharing at low frequency, while the layer leakage inductance controls the current at high frequency. Indeed, the low frequency current can even split between parallel windings with identical geometry (resistance); however, there will be uneven current sharing at high frequencies if the layer inductance is different from each other. Since the layer resistance rarely affects the current sharing at high frequencies, the layer self-resistance in the analytical current sharing model will be represented by its DC form $R_{dc} = \rho l / ch$ for simplification; the mutual resistance is ignored. And the focus now will be shifted to the layer leakage inductance.

The leakage inductance can be derived from the complex magnetic power [20] in the transformer window area as (15), where $L_{self,i}$ is the self-inductance of the i th layer and M_{ij} is the mutual inductance between the i th and j th layer. The analytical complex magnetic power in terms of the magnetic field [21] is expressed as (16).

$$P_M = \frac{1}{2} j\omega \sum_{i=1}^n L_{self,i} I_i^2 + j\omega \sum_{i=1}^n \sum_{j>i}^n M_{ij} I_i I_j \quad (15)$$

$$P_M = 2j\omega \iiint \frac{1}{4} \mu |\mathbf{H}|^2 dv \quad (16)$$

To calculate the layer inductance, assumptions are made, including the 1-D assumptions to simplify the analysis: the magnetic field is linear on the z -axis, with the design consideration, winding thickness is comparable to the skin depth. For an example of the Case 2 2P2S parallel winding in Fig. 3(e), the magnetic field intensity within the copper layer ($0 < z < h$) can be expressed as (17)

$$H_{y,p1}(z) = \frac{I_{p1}}{ch} (h - z)$$

$$H_{y,s1}(z) = \begin{cases} \frac{I_{p1}}{ch} z, & h > z > h_0 \\ \frac{I_{p1} - I_{s1}}{ch} (h - z), & h_0 > z > 0 \end{cases}, H_{y,s1}(h_0) = 0 \quad (17)$$

$$H_{y,s2}(z) = \frac{I_{s2}}{ch} z$$

$$H_{y,p2}(z) = \begin{cases} \frac{I_{s2} - I_{p2}}{ch} z, & h > z > h_0 \\ \frac{I_{s2}}{ch} (h - z), & h_0 > z > 0 \end{cases}, H_{y,p2}(h_0) = 0$$

Combing the complex magnetic power in the copper area and the inter-winding area, the total complex magnetic power in the window area of Case 2 in Fig. 3(e) is

$$P_M = \frac{1}{2} j\omega \left\{ L_w \left(\frac{4}{3} (I_{p1}^2 + I_{s2}^2) + \frac{1}{3} (I_{p2}^2 + I_{s1}^2) - (I_{p2} I_{s2} + I_{p1} I_{s1}) \right) + L_{air} \left(\frac{3}{2} (I_{p1}^2 + I_{s2}^2) + \frac{1}{2} (I_{p2}^2 + I_{s1}^2) - (I_{p2} I_{s2} + I_{p1} I_{s1}) \right) \right\} \quad (18)$$

where $L_w = \frac{\mu_0 h l}{c}$, $L_{air} = \frac{\mu_0 \Delta h l}{c}$. From (15) (18), the leakage inductance matrix of Case 2 in Fig. 3(e) is derived as (19) (20)

$$\begin{bmatrix} v_{lp1} \\ v_{ls1} \\ v_{lp2} \\ v_{ls2} \end{bmatrix} = \begin{bmatrix} L_{self,p1} & M_{p1s1} & 0 & 0 \\ M_{p1s1} & L_{self,s1} & 0 & 0 \\ 0 & 0 & L_{self,p2} & M_{p2s2} \\ 0 & 0 & M_{p2s2} & L_{self,s2} \end{bmatrix} \begin{bmatrix} I_{p1} \\ I_{s1} \\ I_{p2} \\ I_{s2} \end{bmatrix} \quad (19)$$

$$L_{self,p1} = L_{self,s2} = \frac{4}{3} L_w + \frac{3}{2} L_{air}$$

$$L_{self,s1} = L_{self,p2} = \frac{1}{3} L_w + \frac{1}{2} L_{air} \quad (20)$$

$$M_{p1s1} = M_{p2s2} = -\frac{1}{2} (L_w + L_{air})$$

We can notice that the inductance of the parallel branches, P1 and P2, S1 and S2, of Case 2 in Fig. 3(e) is different; P1 and S2 layers are expected to share less current at high frequency as a higher inductance. From (19), (21), and KVL, KCL (21)

$$I_{p1} + I_{p2} = I$$

$$I_{s1} + I_{s2} = I$$

$$v_{lp1} + R_{dc} I_{p1} = v_{lp2} + R_{dc} I_{p2}$$

$$v_{ls1} + R_{dc} I_{s1} = v_{ls2} + R_{dc} I_{s2}$$

(21)

The analytical layer current distribution of the Case 2 2P2S 1:1 transformer in Fig. 3(e) is expressed as

$$\frac{I_{p1}}{I} = \frac{I_{s2}}{I} = \frac{6R_{dc} + j\omega(6L_{air} + 5L_w)}{2(6R_{dc} + j\omega(9L_{air} + 8L_w))}$$

$$= \frac{6\rho + j\omega\mu_0 h(5h + 6\Delta h)}{12\rho + j\omega\mu_0 2h(8h + 9\Delta h)} \quad (22)$$

$$\frac{I_{p2}}{I} = \frac{I_{s1}}{I} = \frac{6R_{dc} + j\omega(12L_{air} + 11L_w)}{2(6R_{dc} + j\omega(9L_{air} + 8L_w))}$$

$$= \frac{6\rho + j\omega\mu_0 h(11h + 12\Delta h)}{12\rho + j\omega\mu_0 2h(8h + 9\Delta h)} \quad (23)$$

Similarly, the leakage inductance matrix of case 1 2P2S 1:1 transformer in Fig. 3(a) can be derived as (19) (24)

$$\begin{cases} L_{self,p1} = L_{self,p2} = \frac{4}{3} L_w + \frac{3}{2} L_{air} \\ L_{self,s1} = L_{self,s2} = \frac{1}{3} L_w + \frac{1}{2} L_{air} \\ M_{p1s1} = M_{p2s2} = -\frac{1}{2} (L_w + L_{air}) \end{cases} \quad (24)$$

Wherein, the inductance of the parallel branches is identical, thereby ensuring even current sharing. Combining (19), (24), and KVL, KCL (21), the analytical layer current distribution of the Case 1 2P2S 1:1 transformer in Fig. 3 (a) is expressed as

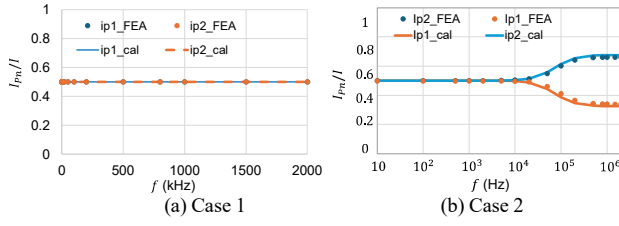
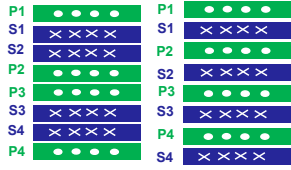


Fig. 10. Analytical & FEA current distribution for 2P2S parallel winding of a 1:1 transformer.

$$\frac{I_{p1}}{I} = \frac{I_{p2}}{I} = \frac{I_{s1}}{I} = \frac{I_{s2}}{I} = \frac{1}{2} \quad (25)$$

Fig. 10 shows the analytical current sharing of parallel windings in Fig. 3 versus the FEA simulated results with $h = 3\text{oz}$. For Case 1 in Fig. 3(a), the current always evenly distributes among parallel layers since the identical layer impedances. For Case 2 in Fig. 3(e), current remains even among parallel layers in the low frequency range where the resistance effect is dominant, while the current tends towards the inner layers (P2 and S1) when frequency increases due to a lower layer inductance of such layers.



(a) Case 1 (b) Case 2
Fig. 11. Interleaved parallel 4P4S winding of a 1:1 transformer

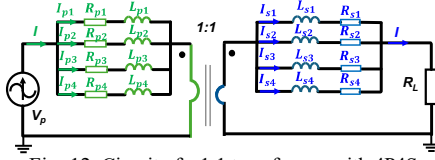


Fig. 12. Circuit of a 1:1 transformer with 4P4S parallel windings.

$$\begin{bmatrix} V_{Lp1} \\ V_{Ls1} \\ V_{Lp2} \\ V_{Ls2} \\ V_{Lp3} \\ V_{Ls3} \\ V_{Lp4} \\ V_{Ls4} \end{bmatrix} = \begin{bmatrix} L_{self,p1} & M_{p1s1} & M_{p1p2} & M_{p1s2} & 0 & 0 & 0 & 0 \\ M_{p1s1} & L_{self,s1} & M_{p2s1} & M_{s1s2} & 0 & 0 & 0 & 0 \\ M_{p1p2} & M_{p2s1} & L_{self,p2} & M_{p2s2} & 0 & 0 & 0 & 0 \\ M_{p1s2} & M_{s1s2} & M_{p2s2} & L_{self,s2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & L_{self,p3} & M_{p3s3} & M_{p3p4} & M_{p3s4} \\ 0 & 0 & 0 & 0 & M_{p3s3} & L_{self,s3} & M_{p4s3} & M_{s3s4} \\ 0 & 0 & 0 & 0 & M_{p3p4} & M_{p4s3} & L_{self,p4} & M_{p4s4} \\ 0 & 0 & 0 & 0 & M_{p3s4} & M_{s3s4} & M_{p4s4} & L_{self,s4} \end{bmatrix} \begin{bmatrix} I_{p1} \\ I_{s1} \\ I_{p2} \\ I_{s2} \\ I_{p3} \\ I_{s3} \\ I_{p4} \\ I_{s4} \end{bmatrix} \quad (26)$$

$$\begin{aligned} L_{self,p1} &= L_{self,s4} = \frac{10}{3}L_w + \frac{7}{2}L_{air}, & L_{self,s1} &= L_{self,p4} = \frac{7}{3}L_w + \frac{5}{2}L_{air} \\ L_{self,p2} &= L_{self,s3} = \frac{4}{3}L_w + \frac{3}{2}L_{air}, & L_{self,s2} &= L_{self,p3} = \frac{1}{3}L_w + \frac{1}{2}L_{air} \\ M_{p1s1} &= M_{p4s4} = -\frac{5}{2}(L_w + L_{air}), & M_{p1p2} &= M_{s3s4} = \frac{3}{2}(L_w + L_{air}) \\ M_{p1s2} &= M_{p3s4} = M_{p2s2} = M_{p3s3} = -\frac{1}{2}(L_w + L_{air}) \\ M_{s1s2} &= M_{p3p4} = \frac{1}{2}(L_w + L_{air}) \end{aligned} \quad (27)$$

Some similarities can be found from the derived inductance matrix of 4P4S parallel windings in (26) (27) and the inductance matrix of 2P2S parallel windings in (19) (20). The inductance of the 2P2S parallel winding configuration is equivalent to that of the inner 2P2S section of the 4P4S winding, indicating that their

magnetic and geometric parameters are closely matched. Based on this concept, the inductance matrix of a 1:1 transformer with an n -layer of primaries and an n -layer of secondaries ($nPnS$) in parallel can be derived as

$$L_{self,i} = \left(\frac{1}{3} + (i-1)\right)L_w + \left(\frac{1}{2} + (i-1)\right)L_{air} \quad (i \leq n) \quad (28)$$

Where i is the number of layers starting from the n th layer. The mutual inductance between a primary layer and a secondary layer or two parallel layers can be expressed as

$$M_{ij} = \begin{cases} -\left(\frac{1}{2} + (i-1)\right)(L_{air} + L_w) & \text{for P and S} \\ \left(\frac{1}{2} + (i-1)\right)(L_{air} + L_w) & \text{for PP or SS} \end{cases} \quad (i, j \leq n, j > i) \quad (29)$$

Therefore, the leakage inductance matrix of Case 1 in Fig. 11(a) can be obtained from (28), (29), and KCL, KVL (30).

$$\begin{aligned} \sum_{i=1}^n I_{pi} &= I, \quad \sum_{i=1}^n I_{si} = I \\ v_{Lpi} + R_{dc}I_{pi} &= v_{Lpj} + R_{dc}I_{pj} \\ v_{Lsi} + R_{dc}I_{si} &= v_{Lsj} + R_{dc}I_{sj} \end{aligned} \quad (30)$$

The equivalent impedance of $nPnS$ windings in parallel with Case 1 configuration for a 1:1 transformer can be derived in (31).

$$\begin{aligned} R_{pi} &= R_{si} = R_{dc} \\ L_{pi} &= -\frac{L_w}{6} \\ L_{si} &= L_{air} + \frac{5}{6}L_w \end{aligned} \quad (31)$$

Since the layer impedance of the $nPnS$ windings with Case 1 configuration is always the same, current can always be equally split among parallel layers as (32)

$$\frac{I_{pi}}{I} = \frac{I_{si}}{I} = \frac{1}{n} \quad (32)$$

In contrast, the layer impedance of $nPnS$ parallel windings with Case 2 configuration for a 1:1 transformer is different from each other; current can be different among the parallel layers. For instance, the current distribution of the 4P4S windings in Fig. 11 (b) is derived from (26), (27), and (30) as (33). Fig. 13 shows its analytical current sharing versus the FEA simulated results with $h=3\text{oz}$.

Given the layer currents solved by the layer impedances, the boundary magnetic field intensity of any layer is known; therefore, the winding loss and resistance can be derived from (12), (13). For example, the analytical ac winding resistance of the 4P4S parallel winding in Fig. 11(b) and the FEA simulated results, with $h = 3\text{oz}$, $\Delta h = 0.127\text{mm}$, $c = 1.8\text{mm}$, core diameter $d = 2.6\text{mm}$, are in Fig. 14. Although currents are not evenly

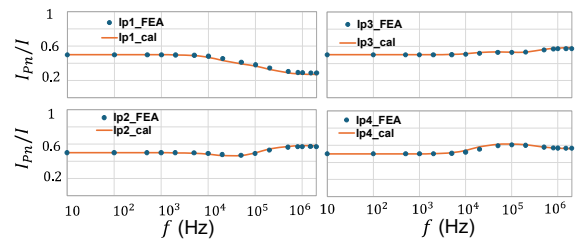


Fig. 13. Analytical & FEA current distribution for Case 2 4P4S parallel winding of a 1:1 transformer.

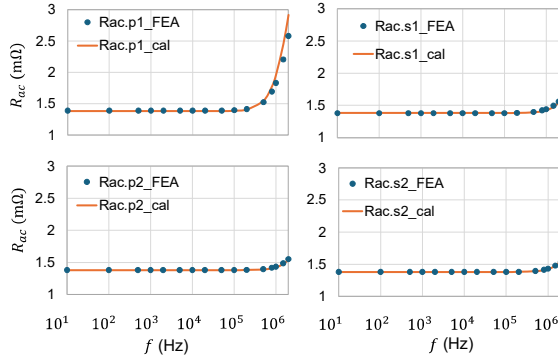


Fig. 14. Analytical & FEA winding resistance of case 2 4P4S parallel winding of a 1:1 transformer.

distributed for Case 2 configuration, the ac winding resistance of the layers shared more current can be lower at high frequency.

To determine an optimal parallel configuration between Case 1 and Case 2, the analytical ac resistance for the 2 cases in Fig. 3 is derived from (12), (13), (22)-(25). For Case 1, the ac resistance of all layers is identical

$$R_{ac,pi} = R_{ac,si} = \frac{P_c}{\frac{1}{2}I_{pi}^2} = R_{dc} \frac{\alpha(\sin(2\alpha) + \sinh(2\alpha))}{\cosh(2\alpha) - \cos(2\alpha)} \quad (34)$$

where $\alpha = h/\delta$. For Case 2, the ac resistance of the topmost and bottommost layer is the same as the ac resistance of Case 1

$$R_{ac,p1} = R_{ac,s2} = R_{dc} \frac{\alpha(\sin(2\alpha) + \sinh(2\alpha))}{\cosh(2\alpha) - \cos(2\alpha)} \quad (35)$$

While for the inner layers, the ac resistance is dependent on the current sharing as in (36).

Fig. 15 shows the layer ac resistance of P1 and P2 of the 2P2S Case 2 in Fig. 3(e). P2 in Case 2 has lower resistance compared to P1 at high frequencies. Since the resistance of P1 in Case 2 is the same as the layer resistance of Case 1, the inner layer winding resistance of Case 2 is always lower than Case 1.

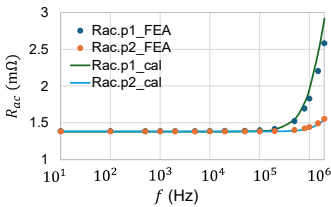


Fig. 15. Analytical & FEA winding resistance of 2P2S Case 2 winding

On the other hand, the current sharing, better in Case 1, also affects the conduction loss. Therefore, the analytical conduction losses, including both current sharing effect and ac resistance effect, for these 2 cases are compared with different winding thickness conditions in Fig.

16. In comparison, Case 2 can achieve the lowest loss from 500kHz ~ 2MHz with a thick copper design due to the less induced eddy current as the green dash line. While for applications from 20kHz~400kHz, Case 1 has lower loss due to a more even current sharing and relatively light eddy current effects in this frequency range as the solid line in Fig. 16. For very low frequency applications, $f < 20$ kHz, both 2 cases can be chosen as a parallel winding design with similar winding loss. Because current sharing is controlled by the layer resistance in this region, where frequency-dependent current redistribution becomes trivial.

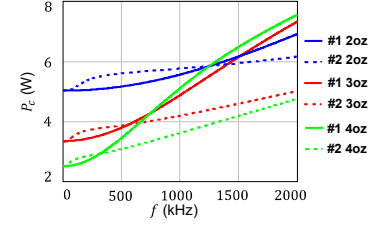


Fig. 16. Analytical conduction loss comparison between 2 cases with interleaved parallel winding.

IV. CONCLUSION

With the increasing demand for power density and efficiency in modern power delivery systems, PCB transformers with parallel windings have become increasingly popular due to their low-profile design and ease of manufacturing. With the development of wide bandgap devices, higher operating frequencies can be employed, and the magnetic size can be further reduced. However, high-frequency operation introduces challenges related to transformer winding losses, including eddy current effects and uneven current sharing between parallel winding layers.

To mitigate eddy current losses at high frequencies, interleaving the primary and secondary windings is commonly adopted, as it reduces the magnitude of the magnetic field penetrating the conductors. Two cases of perfectly interleaved structures are typically considered. Case 1 achieves even current sharing across parallel layers and is widely used. In contrast, Case 2, although it exhibits uneven current sharing, can result in lower eddy current losses at high frequencies due to reduced magnetic field intensity within each layer.

To determine the optimal parallel winding arrangement, this paper presents a simple and effective analytical model based on the layer impedances derived from an equivalent transformer circuit. The model reveals the underlying mechanism of current sharing among parallel windings and is verified using finite element analysis (FEA). It can be extended to arbitrary nP - nS winding configurations and provides a convenient way to solve for individual layer currents.

$$\begin{aligned} \frac{I_{p1}}{I} = \frac{I_{s4}}{I} &= \frac{216R_{dc}^3 + j\omega 108R_{dc}^2(12L_{air} + 11L_w) - \omega^2 18R_{dc}(60L_{air}^2 + 96L_{air}L_w + 37L_w^2) - \omega^3(216L_{air}^3 + 468L_{air}^2L_w + 324L_{air}L_w^2 + 71L_w^3)}{864R_{dc}^3 + j\omega 432R_{dc}^2(15L_{air} + 14L_w) - \omega^2 72R_{dc}(90L_{air}^2 + 150L_{air}L_w + 61L_w^2) - \omega^3 4(378L_{air}^3 + 864L_{air}^2L_w + 639L_w^2 + 152L_w^3)} \\ \frac{I_{p2}}{I} = \frac{I_{s3}}{I} &= \frac{216R_{dc}^3 + j\omega 108R_{dc}^2(14L_{air} + 13L_w) - \omega^2 18R_{dc}(96L_{air}^2 + 164L_{air}L_w + 69L_w^2) - \omega^3(432L_{air}^3 + 1008L_{air}^2L_w + 762L_{air}L_w^2 + 185L_w^3)}{864R_{dc}^3 + j\omega 432R_{dc}^2(15L_{air} + 14L_w) - \omega^2 72R_{dc}(90L_{air}^2 + 150L_{air}L_w + 61L_w^2) - \omega^3 4(378L_{air}^3 + 864L_{air}^2L_w + 639L_w^2 + 152L_w^3)} \\ \frac{I_{p3}}{I} = \frac{I_{s2}}{I} &= \frac{216R_{dc}^3 + j\omega 108R_{dc}^2(16L_{air} + 15L_w) - \omega^2 18R_{dc}(96L_{air}^2 + 160L_{air}L_w + 65L_w^2) - \omega^3(432L_{air}^3 + 1008L_{air}^2L_w + 768L_{air}L_w^2 + 191L_w^3)}{864R_{dc}^3 + j\omega 432R_{dc}^2(15L_{air} + 14L_w) - \omega^2 72R_{dc}(90L_{air}^2 + 150L_{air}L_w + 61L_w^2) - \omega^3 4(378L_{air}^3 + 864L_{air}^2L_w + 639L_w^2 + 152L_w^3)} \\ \frac{I_{p4}}{I} = \frac{I_{s1}}{I} &= \frac{216R_{dc}^3 + j\omega 108R_{dc}^2(18L_{air} + 17L_w) - \omega^2 18R_{dc}(108L_{air}^2 + 180L_{air}L_w + 73L_w^2) - \omega^3(432L_{air}^3 + 972L_{air}^2L_w + 702L_{air}L_w^2 + 161L_w^3)}{864R_{dc}^3 + j\omega 432R_{dc}^2(15L_{air} + 14L_w) - \omega^2 72R_{dc}(90L_{air}^2 + 150L_{air}L_w + 61L_w^2) - \omega^3 4(378L_{air}^3 + 864L_{air}^2L_w + 639L_w^2 + 152L_w^3)} \end{aligned} \quad (33)$$

$$R_{ac,s1} = R_{ac,p2} = \frac{R_{dc}\alpha}{\cos(2\alpha) - \cosh(2\alpha)} \left\{ 4 \left(\frac{I_{p1}^2}{I_{s1}^2} - \frac{I_{p1}}{I_{s1}} \right) (\cosh \alpha \sin \alpha + \cos \alpha \sinh \alpha) - \left(\frac{2I_{p1}^2}{I_{s1}^2} - \frac{2I_{p1}}{I_{s1}} + 1 \right) (\sin(2\alpha) + \sinh(2\alpha)) \right\} \quad (36)$$

Using this current-sharing model in conjunction with a 1-D analytical winding loss formulation, the total AC winding loss of the parallel winding is derived. Based on the comparison of losses in the two interleaved cases, Case 2 is preferable at frequencies above 500 kHz, where minimizing eddy current loss outweighs the impact of uneven current distribution. Case 1, on the other hand, offers lower total loss in the medium-frequency range (20 kHz–400 kHz), where even current sharing is more critical. At frequencies below 20 kHz, both configurations exhibit similar losses, as frequency-dependent current redistribution becomes negligible.

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